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Introduction

I was given a Hakko 483 desoldering station that was destined for the rubbish heap. This document describes the revival of a unit which has seen better days.

The 483 has been out of production for many years now and some spares are very difficult to find. I tested the iron to see that it was still functional – there was no point continuing if the heating element was dead. It was slow to heat up but it did get there, the temperature regulator still worked and it was certainly capable of melting solder. The tip was clear and the glass tube and filters seemed reasonably clean.

Pulling the trigger made the pump run and looking at the tip I could see that melted solder was disappearing into the tube. It seems we had a working unit – or so I thought.

Trying to desolder an IC showed it wasn't working so well. A closer look at the tip showed that the melted solder was oscillating back and forth. Some solder was indeed entering the capture tube but just as much was being spat out onto the bench. This meant the pump was not working correctly.

The pump in the 483 is a big 600mm Hg unit. Most of it is motor but the diaphragm is almost 80mm in diameter. There should be a lot of suction from something if this size. The diaphragm was in good condition with no sign of tearing or perishing. The valve plate was another matter. This is a silicone rubber washer with two small flaps which act as the inlet/outlet valves. The plate had hardened, yellowed and perished. This was the issue.



The only Hakko valve plates I could find was the A-1014 meant for later model soldering stations. I didn't know what size they were but at \$10 for a pair I thought it was a good gamble if I managed to get the station working. As luck would have it they were much too small for the 483. It seemed such a shame to throw out such a well engineered piece of gear and I didn't want to give up just yet.

I wondered if I could make a valve plate. A quick visit to eBay and I found a 500mm² sheet of silicon rubber that was 1mm thick. The price was little over \$10 free post and I thought it was worth another gamble.

The original valve plate was so badly perished that I couldn't use it for a template. I could however look at what was left of it and along with studying the fixing plate and the inside of the pump head work out that I needed to cut two flaps in the new material. I used a hobby knife with a brand new blade and went to it. In less than an hour the job was complete. The job was not very precise but the inlet and outlet ports were covered when required and it looked promising.

The first test was fantastic and I was able to remove some machined pin IC sockets from some 30 year old plated through boards without re-flowing the solder first. A good sign for sure. The hardest part was the ground plane but as I had already cleared the rest of the pins all I had to do was re-flow the GND pin and the socket fell out of the board. Once the socket was out it was easy enough to clear out the re-flowed solder in the last hole.

So, for a little under \$30 I was able to revive the unrepairable. A station that was heading for landfill has found a new life in my workshop. This unit is built like a battleship and as long as the heating element and rubber parts can be replaced I am sure I won't need to buy another station.



The moral of the story is that even if the job looks impossible, sometimes you can find a way around a fault or lack of spares. The amateur is resourceful!